

## EXERCISES

1. Label the following statements as true or false.
  - (a) An elementary matrix is always square.
  - (b) The only entries of an elementary matrix are zeros and ones.
  - (c) The  $n \times n$  identity matrix is an elementary matrix.
  - (d) The product of two  $n \times n$  elementary matrices is an elementary matrix.
  - (e) The inverse of an elementary matrix is an elementary matrix.
  - (f) The sum of two  $n \times n$  elementary matrices is an elementary matrix.
  - (g) The transpose of an elementary matrix is an elementary matrix.
  - (h) If  $B$  is a matrix that can be obtained by performing an elementary row operation on a matrix  $A$ , then  $B$  can also be obtained by performing an elementary column operation on  $A$ .
  - (i) If  $B$  is a matrix that can be obtained by performing an elementary row operation on a matrix  $A$ , then  $A$  can be obtained by performing an elementary row operation on  $B$ .

✓ 2. Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \\ 1 & -1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 3 \\ 1 & -2 & 1 \\ 1 & -3 & 1 \end{pmatrix}, \quad \text{and } C = \begin{pmatrix} 1 & 0 & 3 \\ 0 & -2 & -2 \\ 1 & -3 & 1 \end{pmatrix}.$$

Find an elementary operation that transforms  $A$  into  $B$  and an elementary operation that transforms  $B$  into  $C$ . By means of several additional operations, transform  $C$  into  $I_3$ .

3. Use the proof of Theorem 3.2 to obtain the inverse of each of the following elementary matrices.

$$(a) \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad (b) \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (c) \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$$

4. Prove the assertion made on page 149: Any elementary  $n \times n$  matrix can be obtained in at least two ways—either by performing an elementary row operation on  $I_n$  or by performing an elementary column operation on  $I_n$ .
5. Prove that  $E$  is an elementary matrix if and only if  $E^t$  is.
- ✓ 6. Let  $A$  be an  $m \times n$  matrix. Prove that if  $B$  can be obtained from  $A$  by an elementary row [column] operation, then  $B^t$  can be obtained from  $A^t$  by the corresponding elementary column [row] operation.
7. Prove Theorem 3.1.

- ✓~~8~~. Prove that if a matrix  $Q$  can be obtained from a matrix  $P$  by an elementary row operation, then  $P$  can be obtained from  $Q$  by an elementary matrix of the same type. *Hint*: Treat each type of elementary row operation separately.
- ✓9. Prove that any elementary row [column] operation of type 1 can be obtained by a succession of three elementary row [column] operations of type 3 followed by one elementary row [column] operation of type 2.
- ✓10. Prove that any elementary row [column] operation of type 2 can be obtained by *dividing* some row [column] by a nonzero scalar.
- ✓11. Prove that any elementary row [column] operation of type 3 can be obtained by *subtracting* a multiple of some row [column] from another row [column].
- ✓12. Let  $A$  be an  $m \times n$  matrix. Prove that there exists a sequence of elementary row operations of types 1 and 3 that transforms  $A$  into an upper triangular matrix.

### 3.2 THE RANK OF A MATRIX AND MATRIX INVERSES

In this section, we define the *rank* of a matrix. We then use elementary operations to compute the rank of a matrix and a linear transformation. The section concludes with a procedure for computing the inverse of an invertible matrix.

**Definition.** If  $A \in M_{m \times n}(F)$ , we define the **rank** of  $A$ , denoted  $\text{rank}(A)$ , to be the rank of the linear transformation  $L_A: F^n \rightarrow F^m$ .

Many results about the rank of a matrix follow immediately from the corresponding facts about a linear transformation. An important result of this type, which follows from Fact 3 (p. 100) and Corollary 2 to Theorem 2.18 (p. 102), is that *an  $n \times n$  matrix is invertible if and only if its rank is  $n$ .*

Every matrix  $A$  is the matrix representation of the linear transformation  $L_A$  with respect to the appropriate standard ordered bases. Thus the rank of the linear transformation  $L_A$  is the same as the rank of one of its matrix representations, namely,  $A$ . The next theorem extends this fact to any matrix representation of any linear transformation defined on finite-dimensional vector spaces.

**Theorem 3.3.** Let  $T: V \rightarrow W$  be a linear transformation between finite-dimensional vector spaces, and let  $\beta$  and  $\gamma$  be ordered bases for  $V$  and  $W$ , respectively. Then  $\text{rank}(T) = \text{rank}([T]_{\beta}^{\gamma})$ .

*Proof.* This is a restatement of Exercise 20 of Section 2.4. ■

so

$$I - A = \begin{pmatrix} 0.70 & -0.20 & -0.30 \\ -0.10 & 0.60 & -0.10 \\ -0.30 & -0.20 & 0.70 \end{pmatrix} \quad \text{and} \quad (I - A)^{-1} = \begin{pmatrix} 2.0 & 1.0 & 1.0 \\ 0.5 & 2.0 & 0.5 \\ 1.0 & 1.0 & 2.0 \end{pmatrix}.$$

Since  $(I - A)^{-1}$  is nonnegative, we can find a (unique) nonnegative solution to  $(I - A)x = d$  for any demand  $d$ . For example, suppose that there are outside demands for \$30 billion in food, \$20 billion in clothing, and \$10 billion in housing. If we set

$$d = \begin{pmatrix} 30 \\ 20 \\ 10 \end{pmatrix},$$

then

$$x = (I - A)^{-1}d = \begin{pmatrix} 90 \\ 60 \\ 70 \end{pmatrix}.$$

So a gross production of \$90 billion of food, \$60 billion of clothing, and \$70 billion of housing is necessary to meet the required demands.

### EXERCISES

- ✓ 1. Label the following statements as true or false.
  - (a) Any system of linear equations has at least one solution.
  - (b) Any system of linear equations has at most one solution.
  - (c) Any homogeneous system of linear equations has at least one solution.
  - (d) Any system of  $n$  linear equations in  $n$  unknowns has at most one solution.
  - (e) Any system of  $n$  linear equations in  $n$  unknowns has at least one solution.
  - (f) If the homogeneous system corresponding to a given system of linear equations has a solution, then the given system has a solution.
  - (g) If the coefficient matrix of a homogeneous system of  $n$  linear equations in  $n$  unknowns is invertible, then the system has no nonzero solutions.
  - (h) The solution set of any system of  $m$  linear equations in  $n$  unknowns is a subspace of  $\mathbf{F}^n$ .
2. For each of the following homogeneous systems of linear equations, find the dimension of and a basis for the solution set.

$$\begin{array}{ll}
\text{(a)} \quad \begin{array}{l} x_1 + 3x_2 = 0 \\ 2x_1 + 6x_2 = 0 \end{array} & \text{(b)} \quad \begin{array}{l} x_1 + x_2 - x_3 = 0 \\ 4x_1 + x_2 - 2x_3 = 0 \end{array} \\
\text{(c)} \quad \begin{array}{l} x_1 + 2x_2 - x_3 = 0 \\ 2x_1 + x_2 + x_3 = 0 \end{array} & \text{(d)} \quad \begin{array}{l} 2x_1 + x_2 - x_3 = 0 \\ x_1 - x_2 + x_3 = 0 \\ x_1 + 2x_2 - 2x_3 = 0 \end{array} \\
\text{(e)} \quad x_1 + 2x_2 - 3x_3 + x_4 = 0 & \text{(f)} \quad \begin{array}{l} x_1 + 2x_2 = 0 \\ x_1 - x_2 = 0 \end{array} \\
\text{(g)} \quad \begin{array}{l} x_1 + 2x_2 + x_3 + x_4 = 0 \\ x_2 - x_3 + x_4 = 0 \end{array} &
\end{array}$$

3. Using the results of Exercise 2, find all solutions to the following systems.

$$\begin{array}{ll}
\text{(a)} \quad \begin{array}{l} x_1 + 3x_2 = 5 \\ 2x_1 + 6x_2 = 10 \end{array} & \text{(b)} \quad \begin{array}{l} x_1 + x_2 - x_3 = 1 \\ 4x_1 + x_2 - 2x_3 = 3 \end{array} \\
\text{(c)} \quad \begin{array}{l} x_1 + 2x_2 - x_3 = 3 \\ 2x_1 + x_2 + x_3 = 6 \end{array} & \text{(d)} \quad \begin{array}{l} 2x_1 + x_2 - x_3 = 5 \\ x_1 - x_2 + x_3 = 1 \\ x_1 + 2x_2 - 2x_3 = 4 \end{array} \\
\text{(e)} \quad x_1 + 2x_2 - 3x_3 + x_4 = 1 & \text{(f)} \quad \begin{array}{l} x_1 + 2x_2 = 5 \\ x_1 - x_2 = -1 \end{array} \\
\text{(g)} \quad \begin{array}{l} x_1 + 2x_2 + x_3 + x_4 = 1 \\ x_2 - x_3 + x_4 = 1 \end{array} &
\end{array}$$

4. For each system of linear equations with the invertible coefficient matrix  $A$ ,

- (1) Compute  $A^{-1}$ .
- (2) Use  $A^{-1}$  to solve the system.

$$\begin{array}{ll}
\text{(a)} \quad \begin{array}{l} x_1 + 3x_2 = 4 \\ 2x_1 + 5x_2 = 3 \end{array} & \text{(b)} \quad \begin{array}{l} x_1 + 2x_2 - x_3 = 5 \\ x_1 + x_2 + x_3 = 1 \\ 2x_1 - 2x_2 + x_3 = 4 \end{array}
\end{array}$$

5. Give an example of a system of  $n$  linear equations in  $n$  unknowns with infinitely many solutions.

6. Let  $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$  be defined by  $T(a, b, c) = (a + b, 2a - c)$ . Determine  $T^{-1}(1, 11)$ .

- ✓ 7. Determine which of the following systems of linear equations has a solution.

$$\begin{array}{ll}
 \begin{array}{l} x_1 + x_2 - x_3 + 2x_4 = 2 \\ \text{(a)} \quad x_1 + x_2 + 2x_3 = 1 \\ \quad 2x_1 + 2x_2 + x_3 + 2x_4 = 4 \end{array} & \begin{array}{l} x_1 + x_2 - x_3 = 1 \\ \text{(b)} \quad 2x_1 + x_2 + 3x_3 = 2 \end{array} \\
 \\
 \begin{array}{l} x_1 + 2x_2 + 3x_3 = 1 \\ \text{(c)} \quad x_1 + x_2 - x_3 = 0 \\ \quad x_1 + 2x_2 + x_3 = 3 \end{array} & \begin{array}{l} x_1 + x_2 + 3x_3 - x_4 = 0 \\ \text{(d)} \quad x_1 + x_2 + x_3 + x_4 = 1 \\ \quad x_1 - 2x_2 + x_3 - x_4 = 1 \\ \quad 4x_1 + x_2 + 8x_3 - x_4 = 0 \end{array} \\
 \\
 \begin{array}{l} x_1 + 2x_2 - x_3 = 1 \\ \text{(e)} \quad 2x_1 + x_2 + 2x_3 = 3 \\ \quad x_1 - 4x_2 + 7x_3 = 4 \end{array} & 
 \end{array}$$

8. Let  $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$  be defined by  $T(a, b, c) = (a + b, b - 2c, a + 2c)$ . For each vector  $v$  in  $\mathbb{R}^3$ , determine whether  $v \in R(T)$ .

$$\begin{array}{ll}
 \text{(a)} \quad v = (1, 3, -2) & \text{(b)} \quad v = (2, 1, 1)
 \end{array}$$

9. Prove that the system of linear equations  $Ax = b$  has a solution if and only if  $b \in R(L_A)$ .

- ✓ 10. Prove or give a counterexample to the following statement: If the coefficient matrix of a system of  $m$  linear equations in  $n$  unknowns has rank  $m$ , then the system has a solution.

11. In the closed model of Leontief with food, clothing, and housing as the basic industries, suppose that the input-output matrix is

$$A = \begin{pmatrix} \frac{7}{16} & \frac{1}{2} & \frac{3}{16} \\ \frac{5}{16} & \frac{1}{6} & \frac{5}{16} \\ \frac{1}{4} & \frac{1}{3} & \frac{1}{2} \end{pmatrix}.$$

At what ratio must the farmer, tailor, and carpenter produce in order for equilibrium to be attained?

12. A certain economy consists of two sectors: goods and services. Suppose that 60% of all goods and 30% of all services are used in the production of goods. What proportion of the total economic output is used in the production of goods?

13. In the notation of the open model of Leontief, suppose that

$$A = \begin{pmatrix} \frac{1}{2} & \frac{1}{5} \\ \frac{1}{3} & \frac{1}{5} \end{pmatrix} \quad \text{and} \quad d = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$$

are the input-output matrix and the demand vector, respectively. How much of each commodity must be produced to satisfy this demand?

To extend  $S$  to a basis for  $V$ , we first obtain a basis  $\beta$  for  $V$ . To do so, we solve the system of linear equations that defines  $V$ . Since in this case  $V$  is defined by a single equation, we need only write the equation as

$$x_1 = -7x_2 - 5x_3 + 4x_4 - 2x_5$$

and assign parametric values to  $x_2$ ,  $x_3$ ,  $x_4$ , and  $x_5$ . If  $x_2 = t_1$ ,  $x_3 = t_2$ ,  $x_4 = t_3$ , and  $x_5 = t_4$ , then the vectors in  $V$  have the form

$$\begin{aligned}(x_1, x_2, x_3, x_4, x_5) &= (-7t_1 - 5t_2 + 4t_3 - 2t_4, t_1, t_2, t_3, t_4) \\ &= t_1(-7, 1, 0, 0, 0) + t_2(-5, 0, 1, 0, 0) + t_3(4, 0, 0, 1, 0) + t_4(-2, 0, 0, 0, 1).\end{aligned}$$

Hence

$$\beta = \{(-7, 1, 0, 0, 0), (-5, 0, 1, 0, 0), (4, 0, 0, 1, 0), (-2, 0, 0, 0, 1)\}$$

is a basis for  $V$  by Theorem 3.15.

The matrix whose columns consist of the vectors in  $S$  followed by those in  $\beta$  is

$$\begin{pmatrix} -2 & 1 & -5 & -7 & -5 & 4 & -2 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 & 1 & 0 & 0 \\ -1 & -1 & 1 & 0 & 0 & 1 & 0 \\ -1 & -1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix},$$

and its reduced row echelon form is

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 0 & -1 \\ 0 & 1 & 0 & 0 & -.5 & 0 & 0 \\ 0 & 0 & 1 & 1 & .5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Thus

$$\{(-2, 0, 0, -1, -1), (1, 1, -2, -1, -1), (-5, 1, 0, 1, 1), (4, 0, 0, 1, 0)\}$$

is a basis for  $V$  containing  $S$ .  $\blacklozenge$

### EXERCISES



1. Label the following statements as true or false.

- (a) If  $(A'|b')$  is obtained from  $(A|b)$  by a finite sequence of elementary column operations, then the systems  $Ax = b$  and  $A'x = b'$  are equivalent.

- (b) If  $(A'|b')$  is obtained from  $(A|b)$  by a finite sequence of elementary row operations, then the systems  $Ax = b$  and  $A'x = b'$  are equivalent.
- (c) If  $A$  is an  $n \times n$  matrix with rank  $n$ , then the reduced row echelon form of  $A$  is  $I_n$ .
- (d) Any matrix can be put in reduced row echelon form by means of a finite sequence of elementary row operations.
- (e) If  $(A|b)$  is in reduced row echelon form, then the system  $Ax = b$  is consistent.
- (f) Let  $Ax = b$  be a system of  $m$  linear equations in  $n$  unknowns for which the augmented matrix is in reduced row echelon form. If this system is consistent, then the dimension of the solution set of  $Ax = 0$  is  $n - r$ , where  $r$  equals the number of nonzero rows in  $A$ .
- (g) If a matrix  $A$  is transformed by elementary row operations into a matrix  $A'$  in reduced row echelon form, then the number of nonzero rows in  $A'$  equals the rank of  $A$ .

✓ 2. Use Gaussian elimination to solve the following systems of linear equations.

- |                                                                                                                                                                                                                                        |                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>(a) <math display="block">\begin{aligned} x_1 + 2x_2 - x_3 &amp;= -1 \\ 2x_1 + 2x_2 + x_3 &amp;= 1 \\ 3x_1 + 5x_2 - 2x_3 &amp;= -1 \end{aligned}</math></p>                                                                         | <p>(b) <math display="block">\begin{aligned} x_1 - 2x_2 - x_3 &amp;= 1 \\ 2x_1 - 3x_2 + x_3 &amp;= 6 \\ 3x_1 - 5x_2 &amp;= 7 \\ x_1 + 5x_3 &amp;= 9 \end{aligned}</math></p> |
| <p>(c) <math display="block">\begin{aligned} x_1 + 2x_2 + 2x_4 &amp;= 6 \\ 3x_1 + 5x_2 - x_3 + 6x_4 &amp;= 17 \\ 2x_1 + 4x_2 + x_3 + 2x_4 &amp;= 12 \\ 2x_1 - 7x_3 + 11x_4 &amp;= 7 \end{aligned}</math></p>                           |                                                                                                                                                                              |
| <p>(d) <math display="block">\begin{aligned} x_1 - x_2 - 2x_3 + 3x_4 &amp;= -7 \\ 2x_1 - x_2 + 6x_3 + 6x_4 &amp;= -2 \\ -2x_1 + x_2 - 4x_3 - 3x_4 &amp;= 0 \\ 3x_1 - 2x_2 + 9x_3 + 10x_4 &amp;= -5 \end{aligned}</math></p>            |                                                                                                                                                                              |
| <p>(e) <math display="block">\begin{aligned} x_1 - 4x_2 - x_3 + x_4 &amp;= 3 \\ 2x_1 - 8x_2 + x_3 - 4x_4 &amp;= 9 \\ -x_1 + 4x_2 - 2x_3 + 5x_4 &amp;= -6 \end{aligned}</math></p>                                                      | <p>(f) <math display="block">\begin{aligned} x_1 + 2x_2 - x_3 + 3x_4 &amp;= 2 \\ 2x_1 + 4x_2 - x_3 + 6x_4 &amp;= 5 \\ x_2 + 2x_4 &amp;= 3 \end{aligned}</math></p>           |
| <p>(g) <math display="block">\begin{aligned} 2x_1 - 2x_2 - x_3 + 6x_4 - 2x_5 &amp;= 1 \\ x_1 - x_2 + x_3 + 2x_4 - x_5 &amp;= 2 \\ 4x_1 - 4x_2 + 5x_3 + 7x_4 - x_5 &amp;= 6 \end{aligned}</math></p>                                    |                                                                                                                                                                              |
| <p>(h) <math display="block">\begin{aligned} 3x_1 - x_2 + x_3 - x_4 + 2x_5 &amp;= 5 \\ x_1 - x_2 - x_3 - 2x_4 - x_5 &amp;= 2 \\ 5x_1 - 2x_2 + x_3 - 3x_4 + 3x_5 &amp;= 10 \\ 2x_1 - x_2 - 2x_4 + x_5 &amp;= 5 \end{aligned}</math></p> |                                                                                                                                                                              |

$$\begin{array}{lcl}
 & 3x_1 - x_2 + 2x_3 + 4x_4 + x_5 = & 2 \\
 \text{(i)} & x_1 - x_2 + 2x_3 + 3x_4 + x_5 = & -1 \\
 & 2x_1 - 3x_2 + 6x_3 + 9x_4 + 4x_5 = & -5 \\
 & 7x_1 - 2x_2 + 4x_3 + 8x_4 + x_5 = & 6 \\
 & 2x_1 + 3x_3 - 4x_5 = & 5 \\
 \text{(j)} & 3x_1 - 4x_2 + 8x_3 + 3x_4 = & 8 \\
 & x_1 - x_2 + 2x_3 + x_4 - x_5 = & 2 \\
 & -2x_1 + 5x_2 - 9x_3 - 3x_4 - 5x_5 = & -8
 \end{array}$$

3. Suppose that the augmented matrix of a system  $Ax = b$  is transformed into a matrix  $(A'|b')$  in reduced row echelon form by a finite sequence of elementary row operations.
- (a) Prove that  $\text{rank}(A') \neq \text{rank}(A'|b')$  if and only if  $(A'|b')$  contains a row in which the only nonzero entry lies in the last column.
- (b) Deduce that  $Ax = b$  is consistent if and only if  $(A'|b')$  contains no row in which the only nonzero entry lies in the last column.
4. For each of the systems that follow, apply Exercise 3 to determine whether the system is consistent. If the system is consistent, find all solutions. Finally, find a basis for the solution set of the corresponding homogeneous system.

$$\begin{array}{lcl}
 & x_1 + 2x_2 - x_3 + x_4 = 2 & x_1 + x_2 - 3x_3 + x_4 = -2 \\
 \text{(a)} & 2x_1 + x_2 + x_3 - x_4 = 3 & \text{(b)} \quad x_1 + x_2 + x_3 - x_4 = 2 \\
 & x_1 + 2x_2 - 3x_3 + 2x_4 = 2 & x_1 + x_2 - x_3 = 0 \\
 & x_1 + x_2 - 3x_3 + x_4 = 1 & \\
 \text{(c)} & x_1 + x_2 + x_3 - x_4 = 2 & \\
 & x_1 + x_2 - x_3 = 0 &
 \end{array}$$

- \* ✓ 5. Let the reduced row echelon form of  $A$  be

$$\begin{pmatrix} 1 & 0 & 2 & 0 & -2 \\ 0 & 1 & -5 & 0 & -3 \\ 0 & 0 & 0 & 1 & 6 \end{pmatrix}.$$

Determine  $A$  if the first, second, and fourth columns of  $A$  are

$$\begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix},$$

respectively.

6. Let the reduced row echelon form of  $A$  be

$$\begin{pmatrix} 1 & -3 & 0 & 4 & 0 & 5 \\ 0 & 0 & 1 & 3 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$